

Data Overview

Total column's

- State
- District
- Year
- Month
- Rainfall

Calculated column's

- Season
- Avg Annual Rainfall (District)
- Rainfall Category

Calculated Measure's

- Total Rainfall
- Monsoon Total
- PostMonsoon Rainfall
- Monsoon Share %

Deep information

Calculated column's

1.Season =

```
SWITCH(  
    TRUE(),  
    'india_monthly_rainfall_data'[Month] IN { "Mar", "Apr", "May" }, "Summer",  
    'india_monthly_rainfall_data'[Month] IN { "Jun", "Jul", "Aug", "Sep" }, "Monsoon",  
    'india_monthly_rainfall_data'[Month] IN { "Oct", "Nov" }, "PostMonsoon",  
    'india_monthly_rainfall_data'[Month] IN { "Dec", "Jan", "Feb" }, "Winter",  
    "Other"  
)
```

This DAX formula assigns each month in the rainfall dataset to its respective Indian season. It uses the SWITCH(TRUE()) pattern to check conditions: March to May are labeled as **Summer**, June to September as **Monsoon**, October and November as **PostMonsoon**, and December to February as **Winter**. If a month doesn't fall into these groups, it is marked as **Other**. This helps in analyzing rainfall data season-wise instead of just by individual months.

2.Avg Annual Rainfall (District) =

```
AVERAGEX(  
    VALUES('india_monthly_rainfall_data'[District]),  
    CALCULATE(  
        SUMX(  
            'india_monthly_rainfall_data',  
            VALUE('india_monthly_rainfall_data'[Rainfall])  
        )  
    )  
)
```

VALUES('india_monthly_rainfall_data'[District]) takes the unique list of districts.

For each district, CALCULATE(SUMX(...)) adds up all the rainfall values across months, giving the total annual rainfall for that district.

Finally, AVERAGEX(...) averages those annual totals across all districts.

3.Rainfall Category =

```
SWITCH(  
    TRUE(),  
    'india_monthly_rainfall_data'[Rainfall] = 0, "No Rain",  
    'india_monthly_rainfall_data'[Rainfall] <= 2.5, "Light Rain",  
    'india_monthly_rainfall_data'[Rainfall] <= 7.5, "Moderate Rain",  
    'india_monthly_rainfall_data'[Rainfall] <= 124.4, "Heavy Rain",  
    'india_monthly_rainfall_data'[Rainfall] > 124.4, "Very Heavy / Extreme",  
    "Unknown"  
)
```

This DAX formula creates a **rainfall category** column by classifying rainfall values into ranges.

- If rainfall = **0** → "No Rain"
- If rainfall ≤ **2.5 mm** → "Light Rain"

- If rainfall $\leq 7.5 \text{ mm}$ $\rightarrow$ "Moderate Rain"
- If rainfall $\leq 124.4 \text{ mm}$ $\rightarrow$ "Heavy Rain"
- If rainfall $> 124.4 \text{ mm}$ $\rightarrow$ "Very Heavy / Extreme"
- Otherwise $\rightarrow$ "Unknown"

In short, it converts the numeric rainfall values into meaningful categories for easier analysis.

Calculated Measure's

1.Total Rainfall = SUM('india_monthly_rainfall_data'[Rainfall])

This DAX formula simply calculates the **total rainfall** by adding up all values in the Rainfall column of your dataset.

2.Monsoon Total =

CALCULATE([Total Rainfall], 'india_monthly_rainfall_data'[Season] = "Monsoon")

This DAX formula calculates the total rainfall only for the Monsoon season.

- [Total Rainfall] is your measure that sums rainfall.
- CALCULATE(... , 'india_monthly_rainfall_data'[Season] = "Monsoon") changes the filter so that only rows where **Season = Monsoon** are considered.

3.PostMonsoon Rainfall =

CALCULATE ([Total Rainfall], 'india_monthly_rainfall_data'[Season] = "PostMonsoon")

This formula calculates the **total rainfall during the PostMonsoon season**.

- [Total Rainfall] is your base measure (sum of rainfall).
- CALCULATE(... , 'india_monthly_rainfall_data'[Season] = "PostMonsoon") applies a filter so only rows with **Season = PostMonsoon** are included.

4.Monsoon Share % =

DIVIDE([Monsoon Total],[Total Rainfall],0)

This DAX formula calculates the **percentage share of monsoon rainfall** out of the total rainfall.

- [Monsoon Total] → rainfall only in monsoon season.
- [Total Rainfall] → overall rainfall.
- DIVIDE(..., 0) → safely divides monsoon rainfall by total rainfall, returning **0** if the denominator is blank or zero.